SOKOINE UNIVERSITY OF AGRICULTURE

School of Engineering and Technology

Course: ABE 721 (AGE 721) - Agricultural Systems Modelling and Simulation

TEST 2

Instructor: Kadeghe Fue, PhD, P.Eng(T)
Time Allowed: 3 Hours

Instructions to Candidates:

- This is an **OPEN-BOOK** examination.
- **NO ELECTRONIC DEVICES** (calculators, laptops, smartphones) are permitted.
- Do not recite definitions from the notes. Questions require the synthesis of theory to solve novel agricultural engineering scenarios.
- For questions requiring algorithmic implementation, you are expected to write the structural logic of the Python scripts by hand in your booklet. Syntax perfection is secondary to logical accuracy and algorithmic flow.
- Answer **ALL** questions. Show all logical steps in your mathematical derivations.
- Total marks: 100.

Section A: System Conceptualization and Mathematical Formulation (35 Marks)

Question 1: Multi-Paradigm Modelling of an Aquaponics Facility (15 Marks)
You are modelling a transient, closed-loop commercial aquaponics system. Fish waste produces toxic Ammonia (A), which a biofilter converts to Nitrate (N). The Nitrate is subsequently transported via water flow ($\dot{Q}$) to a hydroponic bed where it is absorbed by a crop biomass (B).

a) Define your system boundary. If you are predicting the final crop yield (B) at day 60, rigorously justify whether a Lumped (ODE) or Distributed (PDE) spatial domain is required for the hydroponic bed component, assuming the flow rate $\dot{Q}$ is highly turbulent and rapidly mixing. (4 Marks)

b) Derive the coupled mass-balance Ordinary Differential Equations for the transient state of Nitrate mass (N) and Crop Biomass (B). Treat the conversion of Ammonia to Nitrate as an exogenous forcing function $U(t)$, and assume crop uptake follows a linear kinetic rate proportional to both N and B. (6 Marks)

c) **Python Implementation:** Write a Python function `def aquaponics_derivatives(state, t, params):` that takes the state vector `[N, B]`, time `t`, and a parameter dictionary, and returns the mathematically derived derivatives $\left[\frac{dN}{dt}, \frac{dB}{dt}\right]$ to be passed into a numerical solver like `scipy.integrate`. (5 Marks)

Question 2: Spatial Discretization in Thermal Soil Sterilization (20 Marks)
To combat soil-borne pathogens without chemicals, a farm implements a continuous-flow microwave sterilization unit. Soil moves on a conveyor belt while microwaves inject severe heat. The 1D transient heat diffusion within the soil profile is modelled by the PDE: $\frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial z^2} + S(z)$, where $S(z)$ is the microwave energy source term.

a) Formulate the algebraic Explicit Finite Difference Method (FDM) scheme for this specific PDE. Let i represent the spatial node and n represent the time step. Clearly isolate the unknown future state T_i^{n+1}. (6 Marks)

b) The operational speed of the conveyor belt is suddenly doubled, requiring the engineer to double the spatial resolution (halving Δz) of the simulation to capture sudden temperature spikes. Mathematically analyze the catastrophic impact this design change will have on the computational simulation time if the explicit scheme is maintained, referencing the CFL stability condition. (6 Marks)

c) **Python Implementation:** Write a Python function `def explicit_fdm_step(T_current, alpha, dz, dt, S_array):` that executes a single time-step update for the entire spatial array of temperatures (ignoring the boundaries for simplicity). Avoid using `for` loops if possible by utilizing NumPy vector/slice logic, but a logical `for` loop is acceptable. (8 Marks)

Section B: Numerical Execution and Stochastic Calibration (35 Marks)

Question 3: Metaheuristics in Subsurface Drip Calibration (20 Marks)
You are calibrating the Hazen-Williams friction coefficients for a massive, aging subsurface drip irrigation network. The objective is to match simulated nodal pressures with observed sensor data. The traditional gradient-based Levenberg-Marquardt Algorithm (LMA) is failing due to massive pressure noise caused by transient trapped air pockets in the pipes.

a) Explain the fundamental mathematical reason why gradient-based optimization solvers (like LMA) collapse when the error surface (Cost Function) is highly rugged and non-convex, and why Particle Swarm Optimization (PSO) is structurally immune to this specific mathematical failure. (5 Marks)

b) In aging pipe networks, under-predicting pressure is a minor inefficiency, but over-predicting pressure is catastrophic (causing pipes to burst). Deviating from the standard Sum of Squared Errors (SSE), mathematically derive a custom continuous Cost Function $J(p)$ that exponentially penalizes situations where simulated pressure exceeds observed pressure. (7 Marks)

c) **Python Implementation:** Write a Python function `def asymmetric_pressure_cost(P_sim, P_obs):` that calculates and returns your derived custom Cost Function from part (b). Assume the inputs are equal-length NumPy arrays. (8 Marks)

Question 4: Stochastic Risk in Autonomous Harvesting (15 Marks)
A fleet of autonomous combine harvesters relies on a deterministic model to predict mechanical jamming. The model relies heavily on "Crop Density," which varies wildly across the field.

a) The engineer realizes that deterministic modeling is predicting zero jams, yet the machines are failing in the field. Explain how converting this model into a Stochastic Monte Carlo simulation using a Log-Normal distribution for Crop Density resolves this predictive failure. (5 Marks)

b) You are tasked with analyzing the risk of a "1-in-1000 hour catastrophic jam". Critically evaluate why standard Monte Carlo sampling is computationally unfeasible for this specific task, and mathematically detail how Latin Hypercube Sampling (LHS) ensures this rare tail-event is evaluated. (10 Marks)

Section C: System Dynamics and Eco-Hydrological Stability (30 Marks)

Question 5: Jacobian Stability in Lake Eutrophication (15 Marks)
Agricultural phosphorus (P) runoff into a nearby lake triggers massive algae (A) blooms. The algae eventually die, and their decomposition consumes all dissolved oxygen, killing the lake ecosystem. However, a specific biological zooplankton (Z) eats the algae, acting as a natural control.

a) Draw a conceptual Causal Loop Diagram (CLD) representing this ecosystem. Explicitly label the polarities of the links between P, A, and Z, and identify one Reinforcing (R) loop and one Balancing (B) loop. (5 Marks)

b) You mathematically formulate this system into ODEs and linearize it around an equilibrium point x^* (representing a "Clear Water" state). You calculate the Jacobian Matrix and find the eigenvalues to be: $\lambda_1 = -0.05 + 0.3i$ and $\lambda_2 = -0.05 - 0.3i$. Based strictly on the Stability Theorem, prove whether the lake will naturally recover from a sudden, massive phosphorus dumping event, and describe the physical trajectory of the algae population during this recovery. (10 Marks)

Question 6: SPAC and the Genetic Engineering Paradox (15 Marks)
A new genetically modified (GM) wheat cultivar is engineered for extreme heat tolerance. The biological modification resulted in two distinct physical changes to the crop canopy: 1) The canopy albedo (α) decreased significantly (leaves are much darker). 2) The aerodynamic resistance (r_a) increased significantly (leaves are highly curled, trapping stagnant air).

a) Utilizing the foundational physics of the FAO-56 Penman-Monteith Evapotranspiration equation, mathematically derive the competing effects of these two specific modifications. Does the lower albedo attempt to increase or decrease total ET? Does the higher r_a attempt to increase or decrease total ET? (8 Marks)

b) If this GM crop is planted in an extremely humid, windless environment (where the Vapor Pressure Deficit, $e_s - e_a$, approaches zero), structurally prove which of the two genetic modifications will mathematically dominate the Evapotranspiration rate, and conclude if the crop will consume more or less water than a non-GM crop. (7 Marks)

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